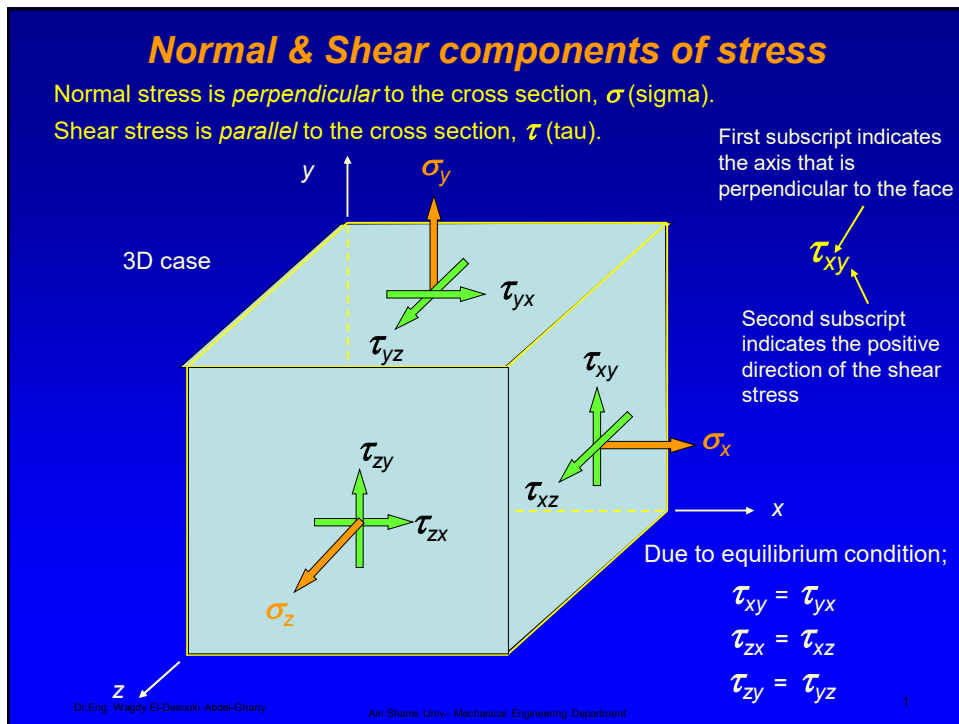
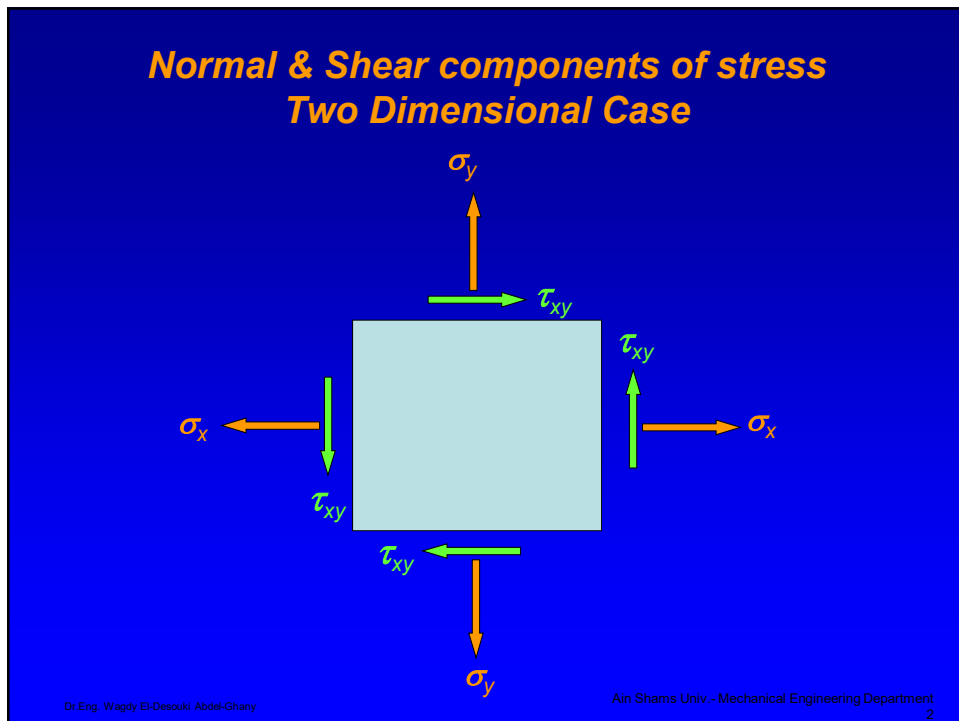




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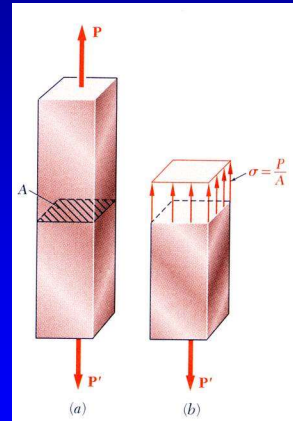


2

Normal Stress Due to Axial Load

A positive sign is used to indicate a tensile stress (tension), a negative sign to indicate a compressive stress (compression)

$$\sigma = \frac{P}{A}$$



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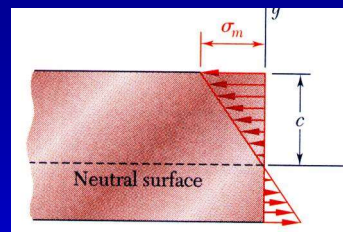
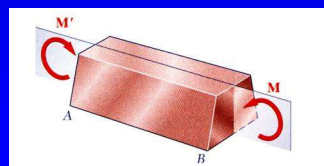
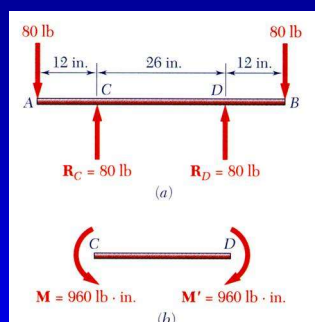
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3

Normal Stress Due to Bending Load

Typical loads on a barbell



Stress distribution

$$\sigma_m = \frac{Mc}{I}$$

Maximum stress at the surface

Where I is area moment of inertia

$I = \pi (d)^4 / 64$ (round cross section)

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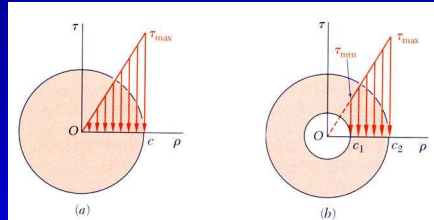
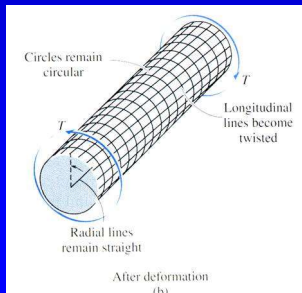
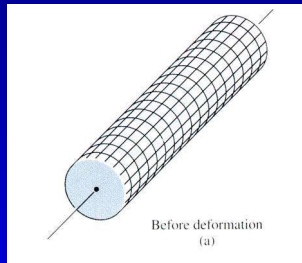
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4

Shear Stress Due to Torque (twisting)

Torsional stress is caused by *twisting* a member



Stress distribution

$$\tau_{\max} = \frac{Tc}{J}$$

Maximum shear stress at the surface

Where J is polar area moment of inertia

$J = \pi (d)^4 / 32$ (round cross section)

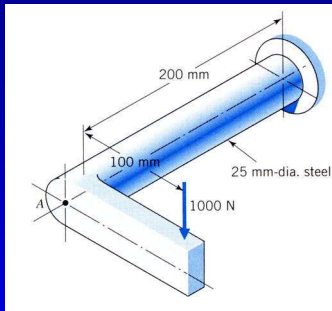
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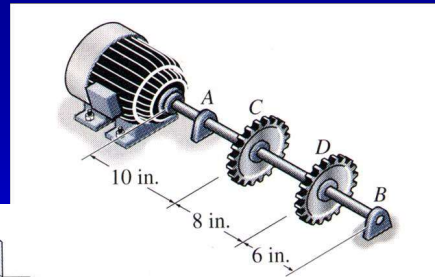
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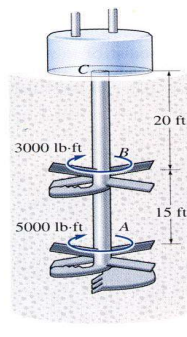
Torsional Stress- examples



Structural member



Power transmission



Mixer

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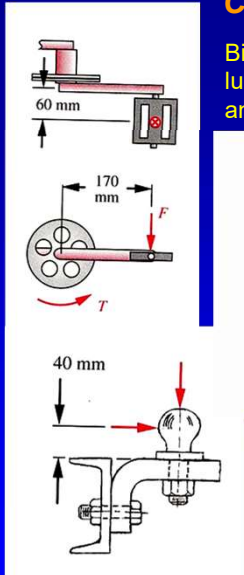
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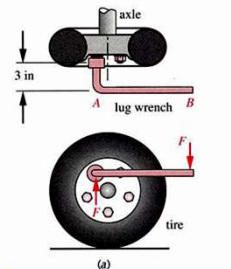
Combined Stresses - examples

Bicycle pedal arm and lug wrench, bending and torsion stresses



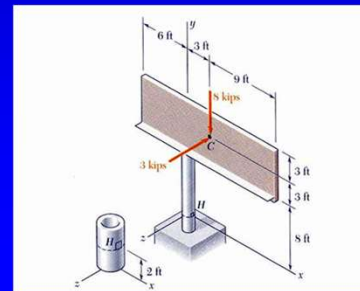
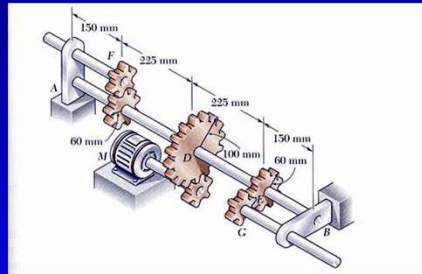
Trailer hitch, bending and axial stresses

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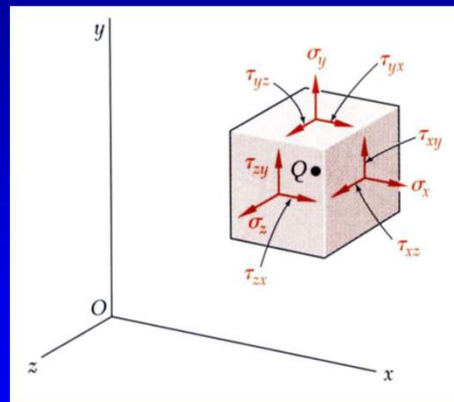


Billboards and traffic signs, bending, axial and torsion stresses

Power transmission, bending and torsion stresses



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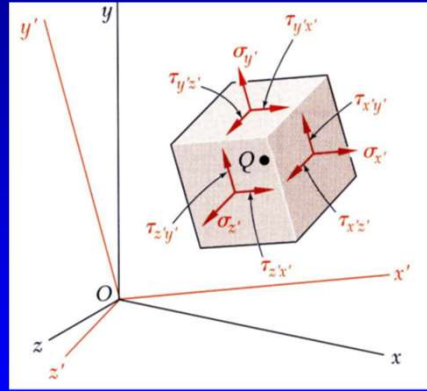


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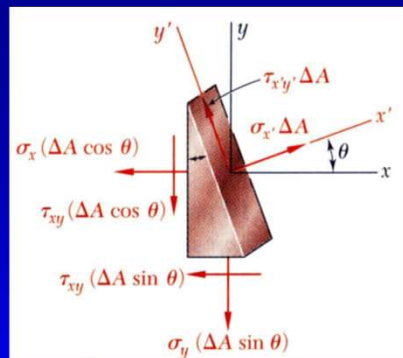
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